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My Health @ EU
eHealth Digital Service Infrastructure
Electronic cross-border health services in the EU

Messaging Profile

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ABSTRACT

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1 Introduction

eHealth DSI utilises a layered communication infrastructure that separates the different concerns and requirements of the legitimate data exchanges. All exposed exchanges are audience restricted through confinement by the eHealth DSI TESTA-ng and mutually authenticate peers as well as communication channels by TLSv1.2/1.3 using a TLS cipher suite in the RSA schema using AES in CBC mode.
eHealth DSI node communication is based on SOAP 1.2 Web Services for business transactions and Syslog protocol (TCP over TLS, defined in RFC5424, RFC5425) for Audit Trail generation.

WSDL definitions of the Web Services are provided by IHE and by the underlying standard (such as ebXML, WS-Trust, etc.). Following the [IHE SOA] model the message structure and payload assignment is defined by the IHE grouping mechanism: from a rather basic SOAP message structure (of the message header and message body), IHE grouping is used to structure the message and encapsulate the different concerns via a *Decorator* Pattern. In eHealth DSI, the **SOAP Header** is governed by WS-Security and SAML (we bear two assertions, the Identity Assertion (IdA) and the Treatment Relationship Confirmation (TRC), as supporting tokens). The **SOAP Body** is defined by the actual standard governing the transaction and it can be based on ebXML for XCA and XDR (for retrieve, fetch, and submit) or on HL7V3 for XCPD queries (for patient discovery). All eHealth DSI transactions are considered to be synchronous (no async option).

1.1 eHealth DSI Secure Messaging Infrastructure

eHealth DSI service providers and consumers use the eHealth DSI messaging infrastructure to exchange request and response messages among each other. The message infrastructure builds upon the eHealth DSI communication infrastructure that connects the eHealth DSI network of trusted nodes (Figure 1).

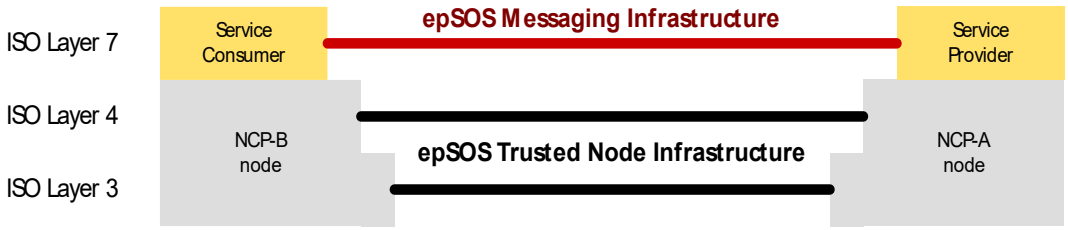


Figure 1: eHealth DSI Trusted Nodes and Messaging infrastructure

The eHealth DSI trusted node infrastructure implements the core eHealth DSI security services that ensure the confidentiality of medical data transmission and the availability and authenticity of eHealth DSI services:

- TESTA-ng,
- message encryption (TLS) and integrity protection, and mutual NCP authentication.

The eHealth DSI messaging infrastructure provides mechanisms for the implementation of the derived eHealth DSI security services (e.g. non-repudiation and access control) and for the standardised enveloping of data and documents:

- transmission of authenticated HP attributes,
- common message format, and
- signature on message elements for auditing and brokering of document authenticity claims.

It must be noted that only NCPs acting as eHealth DSI service providers and service consumers are part of the eHealth DSI trusted node infrastructure as specified in this

document. Points of Care within country B or national data registries/repositories in country A must be connected to the eHealth DSI trusted nodes by means that respect the eHealth DSI end-to-end privacy, security and data protection requirements (see [Security Services Specification] for details).

1.2 Messaging and Transportation Standard Stack

The implementation of the *eHealth DSI Messaging profile* is based on the following standards:

- ebRIM: OASIS/ebXML Registry Information Model v3.0 [OASIS ebRIM 3.0]
- ebRS: OASIS/ebXML Registry Services Specifications v3.0 [OASIS ebRS 3.0]
- XOP: XML-binary Optimized Packaging [W3C XOP]

The following messaging infrastructure security standards MUST be operated mandatorily:

- TESTA-ng/IPSec
- TLS: *Transport Layer Security 1.2 SHALL be used; TLS 1.3 is RECOMMENDED*, with .The choice of cipher suites SHOULD follow [SOG-IS-2020, p 32] to provide mutual authenticated channel
- IHE Audit Trail and Node Authentication [IHE ITI TF-2a] profiling TLS

The following supporting standards are operated on top of the baseline standards as listed above:

- OASIS WS-Security 1.3: with the SAML Token Profile to bear the 2 SAML assertions (Identity and Treatment Confirmation Relationship, TRC) – SAML assertions are considered supporting token (as in not used to secure the SOAP message but the overall transaction)
- SAMLv2: Security Assertion Markup Language Version 2 to encode and bear authentication and authorization claims
- XACMLv2: eXtensible Access Control Policy language Version 2 to encode policies and requests executed both in NCP-B and NCP-A
- XSPA: Cross Enterprise Security and Privacy Authorization Profile for XACMLv2 for Healthcare v1.0 to include healthcare-specific data, such as role, organization, facility type
- HL7-RBAC: HI7 Role Based Access Control Catalogue, to encode the permissions related to the role, on a country-basis
- XUA: IHE Cross Enterprise User Assertion to provide accountability based on the information available in the SAML assertions

1.3 Conventions

The keywords MUST, SHOULD, MAY, SHOULD NOT and MUST NOT are used as defined in [RFC 2119].

eHealth DSI requirements are managed within a central requirements management tool and serialized into [Requirements Catalogue]. References to requirements always use the identifiers as used with the central requirements. Newly defined requirements are handed over to the central requirements management. In case of doubt or conflict, [Requirements Catalogue] MUST be considered to be the normative phrasing of a requirement.

1.4 Terms and Definitions

The country where a patient resides is called "country A" (country of affiliation) in this document. The foreign country where a patient is seeking medical care is called "country B" (country of care) in this document.

The term "Healthcare Professional (HP)" denotes a natural or legal person who provides healthcare services and is allowed to process medical patient data according to the legislation of the country where the healthcare professional is registered. The term "HealthCare Professional Organization (HCPO)" denotes a legal entity that

employs HPs and that is authorized by national law to process and store medical data. A "Point of Care (PoC)" is a single healthcare facility that is operated by an HCPO [eHealth DSI Glossary].
If not explicitly defined otherwise the term "NCP" refers to a gateway which facilitates various aspects of cross-border data sharing (e.g. message forwarding, signature verification). NCPs denote the boundary between the eHealth DSI infrastructure and a country's existing national eHealth infrastructure.

1.5 Status of this Binding

The binding as defined in this document is a normative binding. All eHealth DSI countries MUST implement this binding within their NCP.
Additional or alternative bindings MUST NOT be defined and implemented for the eHealth DSI.

2 Namespaces

XML namespace prefixes are used in this document to stand for their respective namespaces as follows:

Prefix	Namespace
ehdsi	urn:epsos:v1
Soapenv	http://www.w3.org/2003/05/soap-envelope
Wsse	http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-1.0.xsd
Saml	urn:oasis:names:tc:SAML:1.0:assertion
Xacml	urn:oasis:names:tc:xacml:2.0:policy:schema:os
hl7v2	urn:hl7-org:v2
hl7v3	urn:hl7-org:v3
Xds	urn:ihe:iti:xds-b:2007
Xs	http://www.w3.org/2001/XMLSchema
Xsi	http://www.w3.org/2001/XMLSchema-instance
Rimext	urn:ihe:iti:xds-ebrim:extensions:2010[1]
Query	urn:oasis:names:tc:ebxml-regrep:xsd:query:3.0
Rim	urn:oasis:names:tc:ebxml-regrep:xsd:rim:3.0
Rs	urn:oasis:names:tc:ebxml-regrep:xsd:rs:3.0
Lcm	urn:oasis:names:tc:ebxml-regrep:xsd:lcm:3.0

3 eHealth DSI Trusted Node Infrastructure

The mutual trust between a service consumer and a service provider is based on a mutually trusted, secure channel between the underlying network nodes.
The establishment of mutual trust between nodes is performed by:

- TESTA-ng
- Transport Layer Security v1.2[2] [RFC 5426] and v1.3 [RFC 8446]
- IHE Audit Trail and Node Authentication [IHE ITI TF-2a]

[1] IHE will be asked to block this namespace for eHealth DSI extensions in order to avoid conflicts with forthcoming IHE extensions to the ebRIM.
[2] IHE will be asked to block this namespace for eHealth DSI extensions in order to avoid conflicts with forthcoming IHE extensions to the ebRIM.

3.1 TESTA-ng

A gateway-to-gateway TESTA-ng MUST be set up between all eHealth DSI nodes. TESTA-ng/IPSec ESP transport modus MUST be used. Perfect Forwarding Secrecy MUST be activated. SA Lifetime SHOULD be based on the number of exchanged packets and SHOULD NOT exceed 4GB.
Algorithms and key lengths MUST be used acc. [Cryptographic Algorithms]. Gateway certificates MUST comply with the certificate profiles as defined in [X.509 Certificate Profiles]. The issuing CAs and all components and services for managing the lifecycle of the certificates must comply with the respective eHealth DSI security policies (see [Security Services Specification]).

3.2 TLS Configuration

All network nodes running eHealth DSI service consumers or service providers MUST be implemented as IHE *Secure Node* actors acc. to the IHE ATNA profile. The establishment of mutual trust and the setup of the secure transport layer channel between two eHealth DSI nodes are always initiated by a service consumer that connects to a service provider.
The messages for the establishment of the basic transport layer secure channel correspond to the TLS handshake protocol as profiled in the IHE ATNA Integration profile (transaction ITI-19 as specified in section 3.19 of [IHE ITI TF 2a]).
With respect to the ITI-19 transaction specification, the following constraints and extensions apply:

- Algorithms and key lengths MUST be used according [Cryptographic Algorithms].
- The node certificates MUST comply with the eHealth DSI Node Authentication Certificate Profile (see respective node definitions in [X.509 Certificate Profiles] for the certificate profile specifications).
- The issuing CA and all components and services for managing the lifecycle of the eHealth DSI Node Authentication Certificates must comply with the respective eHealth DSI security policies (see [Security Services Specification]).

4 Time Synchronisation

Time synchronisation within the network of eHealth DSI gateways is performed by:

- Network Time Protocol (Version3) [RFC 1305] as described in the
- IHE Consistent Time Integration Profile [IHE ITI TF-2a]

Stratum 2 time servers (*Consistent Time Mediators)* SHOULD by operated by NCPs. All services that rely on consistent time within the eHealth DSI Circle of Trust MUST be operated on a node that acts as a stratum 3 time server (*Consistent Time Consumer)*. This in particular holds for

- services that apply or verify digital signatures on messages, medical data or assertions,
- services that contribute to the security audit trail.

eHealth DSI time synchronisation for *Consistent Time Consumers* is handled by respective mechanisms of the underlying operating systems. The messages exchanged correspond to the NTP transactions described in detail in RFC 1305 and http://www.ntp.org. The underlying network protocol is UDP (port 123). Authentication MAY be enabled with the *ntp authenticate* command.
Consistent Time servers that represent a Stratum n+1 server SHOULD have a configuration with a default polling interval of 4096 seconds at a minimum in order to synchronize the eHealth DSI reference time to all nodes.
Following time servers SHOULD only configure a polling interval of 65536 seconds.
Each Consistent Time Mediator SHOULD accept a maximum clock skew of 256 seconds. With respect to lower the system resources (due to incoming requests) of the Consistent Time Source, Consistent Time Mediators SHOULD peer themselves.

5 eHealth DSI Common Message Format

The eHealth DSI Common Message Format defines the structure and characteristics of the messages exchanged through eHealth DSI; and establishes the preconditions for successful communication. The eHealth DSI Common Message Format describes only the structure of messages as they flow between service consumers and service providers. Messages within NCP's or national infrastructures may use any format desired, and need to be translated to the eHealth DSI Common Message Format before transmission to another NCP.

5.1 Transport Layer Profile

All messages MUST be sent over HTTP 1.1 connections that are layered on top of the eHealth DSI trusted node infrastructure (see section 2).

5.2 Message Layer Profile

The eHealth DSI Common Message Format is a SOAP 1.2 [W3C SOAP 1.2] message contained as the body of an HTPP 1.1 [RFC 2616] message.
All messages MUST be **SOAP Envelopes** with an XML payload in the **SOAP Body**. Optional binary data MUST be carried as Base 64 encoded octets within the XML payload if not otherwise stated for the respective operations. Request messages MUST be sent using an HTTP POST, response messages are carried over the backchannel, i.e. the HTTP response.
The encoding of the containing XML document MUST be set to UTF-8[3].
All eHealth DSI SOAP messages MUST comply with the WS-I Basic Profile 1.1[4] [WSI BP 1.1].
All eHealth DSI SOAP messages MUST comply with the WS-I Basic Security Profile 1.1 [WSI SBP 1.1].
eHealth DSI is currently supporting those SOAP basic types:

- SIMPLE SOAP

Although it is not specifically specified in any baseline standard, eHealth DSI and IHE constrain the request / response preservation by specifying that the response must be provided in the same encoding / wrapping as the request.

5.3 XML Message Schema Format

All eHealth DSI SOAP message MUST be described in a WSDL 1.1 [W3C WSDL 1.1] Service Description.
All WSDL type definitions MUST be in XML Schema format. One Schema must be provided for the request message, and one for the response message.
For better maintainability, an XML Schema import or include statement in the WSDL file SHOULD be used, so the XML Schema can be maintained and reused as a separate entity. If the XML Schema is small, and reuse is not expected, the entire Schema MAY be specified in the WSDL types section (especially in the case of RPC-style transactions where one or a few parameters of rather simple types are used).

5.4 SOAP Binding

For all messages a SOAP 1.2 HTTP Binding MUST be provided in the WSDL.
All SOAP Bindings in the WSDL MUST specify style="document".
All SOAP Bindings in the WSDL MUST specify use="literal".
The naming of the messages MUST be as defined for the used standard.

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[3] UTF-8 is more efficient than UTF-16 for European languages. Older encodings such as ISO-8859-x do not cover all languages in a single encoding, and will only pose interoperability problems. UTF-8 is the default in XML, and coverage is a requirement of the XML specification.

[4] While WSI Basic profile 1.1 does not formally support SOAP 1.2, it takes into consideration SOAP 1.2 by having requirements which are specifically for compatibility with SOAP 1.2. IHE and eHealth DSI plan to consider adoption of WS-I BP 2.0 [WSI BP 2.0] as soon as it is approved by WSI.

The "soapenv:mustUnderstand="1" attribute MUST be set as defined for the used standard.

5.5 Embedding of Security Token

Each SOAP message MUST include a <wsse:Security> section within the SOAP header.

5.5.1 SAML Assertions

Request messages are safeguarded by up to three SAML assertions that attest the authenticity of the user (optional Next of Kin) and the existence of a treatment relationship (See chapter 2 on which assertions are required for each operation).
SAML assertions are contained within the <wsse:Security> section of the SOAP header. The HP Identity Assertion always takes the role of a Supporting Token.
The saml:Advice element MUST be used to define the linkage between the three assertions (see [SAML Profile] on the generic structure of the eHealth DSI TRC, NOK Assertion). All assertions MUST have been issued for the same subject. The relying party MUST verify the correct linkage of the SAML assertions and the match of the <Subject> elements' contents.

5.5.2 Message Signature

To preserve the integrity and authenticity of a message and to attest the NCP origin of the message, elements of each message MAY be signed by the protection token. If WS SecurityPolicy is used, a signed elements assertion MUST be used to refer to the message parts to be signed[5].
The following table defines which token MUST be used as protection token and which elements of the message MUST be covered by the signature.

	Request Message	**Response Message**
Protection Token	X.509 Token of NCP-B	X.509 Token of NCP-A
Signed Elements	/Envelope/Body /Envelope/Header/Security/Assertion	/Envelope/Body

Signatures MUST be placed within a XML-Signature compliant <ds:Signature/> element inside the SOAP security header. The recommendations given in section 8 of [OASIS WS-Security 1.1] SHOULD be considered. In addition the following constraints apply:

Signature Parameter	**Usage Convention**
CanonicalizationMethod	SHOULD be "http://www.w3.org/2001/10/xml-exc-c14n#"
Transformation	Exclusive XML canonicalization SHOULD be used (http://www.w3.org/2001/10/xml-exc-c14n#, acc. [W3C XMLDSig] and [W3C XML-EXC 1.0]). As inclusive namespaces other prefixes than the ones defined in section 2 of this document MUST NOT be used.
SignatureMethod	The signature method MUST comply with the eHealth DSI recommendations on algorithms and key lengths (see [Cryptographic Algorithms]). For signing message elements the signature method - "http://www.w3.org/2001/04/xmldsig-more#rsa-sha256" SHOULD be used.
DigestMethod	The hash algorithm MUST comply with the eHealth DSI recommendations on algorithms and key lengths (see [Cryptographic Algorithms]). For signing message elements the digest method - http://www.w3.org/2001/04/xmlenc#sha256 SHOULD be used.

[5] Only in cases where the framework does not allow for multiple XPath expressions, a signed parts assertion SHOULD be used.

KeyInfo	This element MUST contain a wsse:SecurityTokenReference element which references the protection token.

5.6 Processing of SOAP Messages

A sending service consumer SHOULD add SOAP Message Headers in the order in which the receiving service provider is expected to process them.
A receiving service provider MUST not rely on a specific order of SOAP Message Headers for correct processing.
A receiving service provider MAY rely on a specific order of SOAP Message Headers for faster processing.

6 Exception Handling

In general eHealth DSI distinguishes between four coarse-grained failure situations that MAY require an exchange of respective fault messages between NCPs:

- Communication failures: The message cannot be delivered to the designated service; e.g. because the establishment of a secure communication link failed or a link is broken.
- eHealth DSI Message encoding and consistency failures: The message was received but it cannot be processed; e.g. because the security token is broken or the message does not comply with the eHealth DSI message encoding and security rules.
- Message processing failures: The message was decoded but it cannot be processed; e.g. because of security/privacy reasons or because the request does not match with the states of the affected security and business objects.
- Document encoding failures: The message can be processed but the requested document cannot be completely encoded as specified in [CDA templates].

General system failures are handled based on their implication (e.g. if a message cannot be delivered due to a buffer error, the communication failure mechanism is used to propagate the error; if the same failure occurs during message processing, the message processing failure mechanism is used).
In the following sections it is defined, how failures of the first three kinds are handled by eHealth DSI. Failures of the fourth kind are handled within the documents (e.g. by nullifying fields or using specific error codes; see [CDA templates]).

6.1 Communication Failures

Communication failures are raised by the existing mechanisms of the communication and messaging protocols RequireClientCertificate. They are handled on the layer where they occurred. eHealth DSI does not define new error codes for these kinds of failures and eHealth DSI does not define any requirements for raising and processing these errors that are beyond the pre-setting of the respective standards. This includes that errors of this kind can occur at both communicating gateways and MAY require action to be taken by both gateways.
In case of a communication failure an audit trail entry MUST be written at NCP-B:

eHealth DSI Instance	Opt.	Description
Event	R	Service that was to be called
Requesting Point of Care	R	HCPO that issued the original request.
Human Requestor	R	HP that triggered the request
Source Gateway	R	Service consumer node address at the country of Care
Target Gateway	R	Service provider node that did not respond to the request
Audit Source	X	-
Patient	R	Patient
Event Target	X	-

Error Message	R	Error data as provided by the layer that detected the communication failure

6.2 Encoding and Consistency Failures

Encoding and consistency problems are detected at the protocol terminator or other eHealth DSI-side internal components at the service providing NCP. Errors that origin in an improper encoding of the message (envelope, header) or in an inaccurate use of security objects are covered by the SOAP error mechanism.

6.2.1 SOAP Error Profile

Information on faults that occurred during the processing of a request are placed into a SOAP response message body as SOAP 1.2 faults. The respective data type MUST be instantiated as defined in [W3C SOAP 1.2]. eHealth DSI specific error information is encoded with the following elements:

Element name	Format	Opt.	Content
Code/Subcode/Value	QName	R	eHealth DSI error code; see tables below for the defined values
Reason/Text	QName	R	Description of the error (by default the error code is used as the error description; nevertheless a NCP implementation MAY provide the error condition (see tables below) or even more detailed information on the reason of the failure in this element)
Node	URI	O	URI of the system component that caused the failure or (URI encoded) OID of the object that caused the error. The semantics of this entry MUST be determinable by the error code. By default this element holds the URI of the Service Provider where the error was detected.

Table 1: Usage conventions for SOAP faults

Further details on the error MAY be given in a <detail/> element. The receiver of the fault message MUST NOT process the <detail/> element but SHOULD dump its contents into the respective field of the audit trail entry.

6.2.2 General Message Handling Errors

General message handling faults are detected upon receipt of a message. Usually they are not specific for a certain transaction and usually origin in weaknesses related to the implementation, configuration and operation of the eHealth DSI NCP.
The following table lists all general message handling errors. These errors MUST be handled acc. to the eHealth DSI SOAP error profile (see section 6.2.1).

Condition	Code	Subcode	Action to be taken
The service provider is not able to fulfil the request due to an internal problem	Receiver	Busy	Both NCPs MUST write an audit trail entry. The service requestor SHOULD send the request again.
The service provider cannot write an audit trail entry.	Receiver	Audit Log Failure	The service consumer MUST write an audit trail entry. The service provider MUST write a log entry to the systems log. The system administrator MUST process this failure because it indicates a mis-configuration or software error.

6.2.3 SOAP Message Encoding and Addressing Errors

Message encoding and addressing faults are detected upon receipt of a message or at the protocol terminator. Usually they are not specific for a certain transaction and

usually origin in weaknesses related to the implementation, configuration and operation of the eHealth DSI NCP.
The following table lists all message encoding and addressing errors. These errors MUST be handled acc. to the eHealth DSI SOAP error profile (see section 6.2.1).

Condition	Code	Subcode	Action to be taken
The protocol terminator cannot decode the message because of a schema violation in the SOAP envelope	MustUnderstand or DataEncodingUnknown (depending on the source of the error)	Decoding Failure	No audit trails are written. The service consumer MUST write a log entry to the systems log. The system administrator MUST process this failure because it indicates a mis-configuration or software error.
The protocol terminator cannot validate a message because of an unknown namespace or schema	DataEncodingUnknown	Unknown Schema	No audit trails are written. The service consumer MUST write a log entry to the systems log. The system administrator MUST process this failure because it indicates a mis-configuration or software error.
The protocol terminator cannot process the message because it does not know or not support the requested service.	MustUnderstand or DataEncodingUnknown (depending on the source of the error)	Unknown Transaction	No audit trails are written. The service consumer MUST write a log entry to the systems log. The system administrator MUST process this failure because it indicates a mis-configuration or software error.
The protocol terminator rejects the message because of a version mismatch (e. g. service consumer uses deprecated version of the spec)	MustUnderstand or DataEncodingUnknown (depending on the source of the error)	Version Mismatch	No audit trails are written. The service consumer MUST write a log entry to the systems log. The system administrator MUST process this failure because it indicates a mis-configuration or software error.

6.2.4 Security Header Encoding and Consistency Errors

Security header encoding and consistency errors are detected by the security manager component. Usually they are not specific for a certain transaction and usually origin in weaknesses related to the implementation, configuration and operation of the eHealth DSI NCP.
The following table lists all security header related errors. These errors MUST be handled acc. to the eHealth DSI SOAP error profile (see section 6.2.1).

Condition	Code	Subcode	Action to be taken
The provided HP Identity Assertion does not contain all of the required attributes.	Sender	HP Missing Attributes	Both service consumer and service provider MUST write an audit trail entry. The service consumer SHOULD request a new authentication of the HP.
The provided HP Identity Assertion is not valid or timed out.	Sender	Invalid Security Token	Both service consumer and service provider MUST write an audit trail entry. The service consumer SHOULD request a new HP authentication.
The patient identifier is not valid.	Sender	Unknown Patient	Both service consumer and service provider MUST write an audit trail entry. The HP at the country of care SHOULD identify the patient again,

			establish a new security context and retry the request.
An attesting (message) signature cannot be verified.	Sender or Receiver (depending on the source of the failure)	Invalid NCP Signature	Both service consumer and service provider MUST write an audit trail entry. The service provider MUST write a log entry to the systems log. The system administrator MUST process this failure because it indicates a mis-configuration or software error.
The use of SHA-1 as a digesting method is not allowed.	Sender or Receiver (depending on the source of the failure)	Weak Digest	Both service consumer and service provider MUST write an audit trail entry. The service provider MUST write a log entry to the systems log. The service consumer SHOULD re-issue the request using SHA-2 for digesting (message signature and assertion signatures). The service provider SHOULD use the same digesting method for message signatures as the service consumer.
The requestor provided a Confirmation Assertion which is not accepted by the service provider.	Sender	Weak Authorisation	Both service consumer and service provider MUST write an audit trail entry. The HP SHOULD trigger the issuance of a new TRC assertion by NCP-B and re-issue the request.

6.2.5 Audit Trail Considerations

In case of a general message handling error, NCP-B MUST write a full audit trail including an error section as defined in [Audit Trail Profiles] section eHealth DSI HP Assurance Audit Schema.
NCP-A MUST fill all audit trail information that could be decoded from the request message.
If the requested operation cannot be decoded the Event Identification section of the HP Assurance Audit Schema MUST be used as follows:

Field Name	Value Constraints
EventID	MUST be set to EV(EHDSI-00, "unknown", unknown)
EventActionCode	MUST be set to E (execute).
EventDateTime	Time of the occurrence of the failure
EventOutcomeIndicator	MUST be "4" for temporal or recoverable failures, and "8" for permanent failures

If the HP identity cannot be decoded from the HP Identity Assertion, the Human Requestor section of the HP Assurance Audit Schema MUST be used as follows:

Field Name	Value Constraints
UserID	Subject and issuer MUST be set to "unknown". See [SAML Profile] for the mandatory encoding scheme for user identifiers.
UserName	MUST be set to "unknown"
UserIsRequestor	"true"
RoleIDCode	MUST be omitted.

If the patient identity cannot be decoded from the request, the Patient section of the HP Assurance Audit Schema MUST be used as follows:

Field Name	Value Constraints

ParticipantObjectTypeCode	MUST be "1" (Person)
ParticipantObjectTypeCodeRole	MUST be "1" (Patient)
ParticipantObjectIDTypeCode	EV(2, RFC-3881, "Patient Number")
ParticipantObjectID	MUST be "unknown"

7 References

7.1 Normative References

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[OASIS ebRS 3.0] OASIS ebXML: ebXML Registry Services and Protocols version 3.0, available at http://docs.oasis-open.org/regrep/regrep-rs/v3.0/regrep-rs-3.0-os.pdf

[RFC 1305] Mills, David L., Allen, C.: Network Time Protocol (Version3), Specification, Implementation and Analysis. University of Delaware, March 1992.

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7.2 Non-Normative References

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